

Jestin Andrew Arroyo Castillo

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EDUCATION

The University of Texas at Austin

B.S. Mechanical Engineering | Minor: Business Foundations

August 2024 - December 2027

GPA: 3.85

EXPERIENCE

Applied AI Engineering Intern – Agentic Innovations

May – August 2026

- Developed an international supply chain tracker to manage the network between manufacturers and engineering teams
- Scrapes websites for supplier data and automatically refreshes to track changes in prices, lead times, & reliability
- Automates bill of materials (BOMs) tracking and component sourcing to refine budget estimations
- Maps component dependencies and routes optimal workflow patterns based on critical components and slack timeframes
- Generates reports, next courses of action, and alternative parts and suppliers based on team priorities
- Optimized Baja SAE racecar budget by 10.2% and reduced critical component procurement lead times by 10 days

ENGINEERING PROJECTS

EMG-Actuated 6-Axis Bionic Limb – Longhorn Neurotech

- Developing a desktop 6-axis prosthetic arm capable of hand gestures, actuated by an EMG-reading Mudra Link armband
- Translating human movement via actuating three arm segments between the robotic base, mock shoulder, elbow, and wrist
- Executing failure mode and effects analysis (FMEA) to identify and reinforce structural stress points while optimizing weight
- Performing iterative fabrication to prevent premature mechanical failure from material yielding or cyclic fatigue

Baja Vehicle Pedal Box Optimization – Longhorn Baja Racing, Baja SAE

- Applied human-centered design to produce an adjustable set of foot pedals to fit 4 drivers of sharp height differences
- Conducted Finite Element Analysis (FEA) to identify and reinforce weak points and reduce weight by 8.6%
- Implemented risk management frameworks to evaluate failure modes under cyclic loading for a 4-hour endurance race
- Created 3 iterated designs to optimize driver ergonomics, responsiveness, and overall performance

Functional Arm Prosthetic – e-NABLE @ UT Austin

- Designed and fabricated a custom 3D-printed upper-limb prosthetic for a child with an above-the-elbow amputation
- Engineered a durable and functional gripping mechanism capable of supporting objects weighing up to 5 pounds
- Combined CAD modeling and biomechanics principles to replicate natural human movement and joint behavior

LEADERSHIP EXPERIENCE

Prosthetic Mechanical Lead – Longhorn Neurotech

September 2026 – Present

- Leading a team of 10 undergraduates to design and fabricate a 6-axis bionic limb, actuated via an EMG-reading armband
- Directing integration with electrical, software, and research teams to streamline workflows and ensure seamless collaboration
- Upskilling members in mechanical design, structural failure analysis, and thermal dissipation solutions

Vice-Captain – Longhorn Baja Racing, Baja SAE

February 2026 – Present

- Directing administrative operations for a 130+ member engineering organization spanning across 3 main teams
- Driving 7 interdisciplinary sub-teams to refine and optimize integration, achieving 106% membership growth within 2 weeks
- Finished the 4-hour endurance race without major component failure and moved up from 69th to place at 67th as a first-year team

Test Hands Team Lead – e-NABLE @ UT Austin

October 2025 – October 2026

- Guiding a team of 15 undergraduates to build 15 customized, durable, and functional wrist and hand prosthetics
- Partnering with 2 international chapters to distribute each prosthetic device to clients of all ages across the globe
- Mentoring new members in CAD and slicing software to build proficiency and foster team integration

Ergonomics Lead, Co-Founder – Longhorn Baja Racing, Baja SAE

May 2025 – Present

- Coordinating a team of 14 undergraduates to enhance an off-road racecar for safety and driver comfort
- Ensuring our 4 drivers perform comfortably and consistently despite unpredictable track terrain and conditions
- Leading the development of a customizable pedal box and human analog models within chassis and dynamics constraints

SKILLS & CERTIFICATES

- Computer-Aided Design (CAD) – SolidWorks, SolidWorks Simulations, Fusion 360, OnShape, FEA, FMEA
- Design Methods – Human-centered design, biomechanics, risk assessment, tolerancing, GD&T
- Fabrication – FDM 3D printing, additive manufacturing, iterative prototyping, fit testing, manual mill
- Programming & Data Analysis – Python, MATLAB, Microsoft Excel
- Certifications – Clinical Medical Assistant, EKG Technician, Python Programmer